

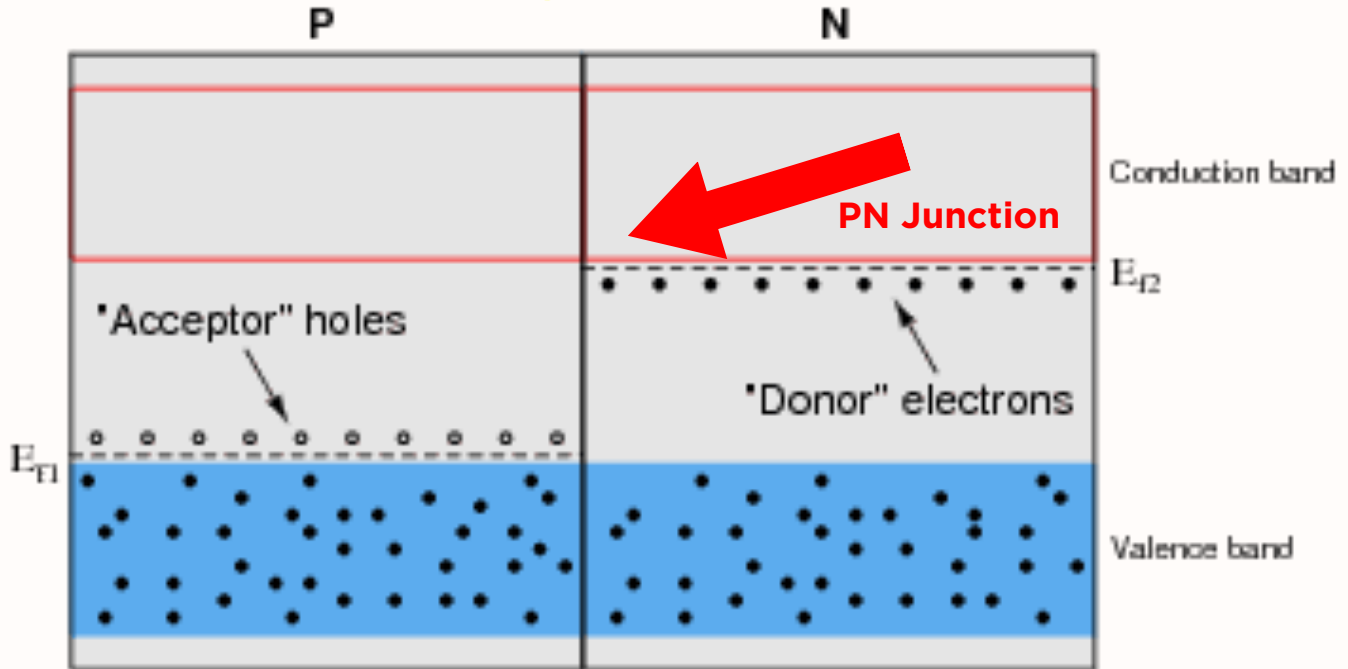


EGCI 232 Engineering Electronics

- Physics of Elements

Basic Structure of electronic device

PN Junction - at 0 K (absolute zero)

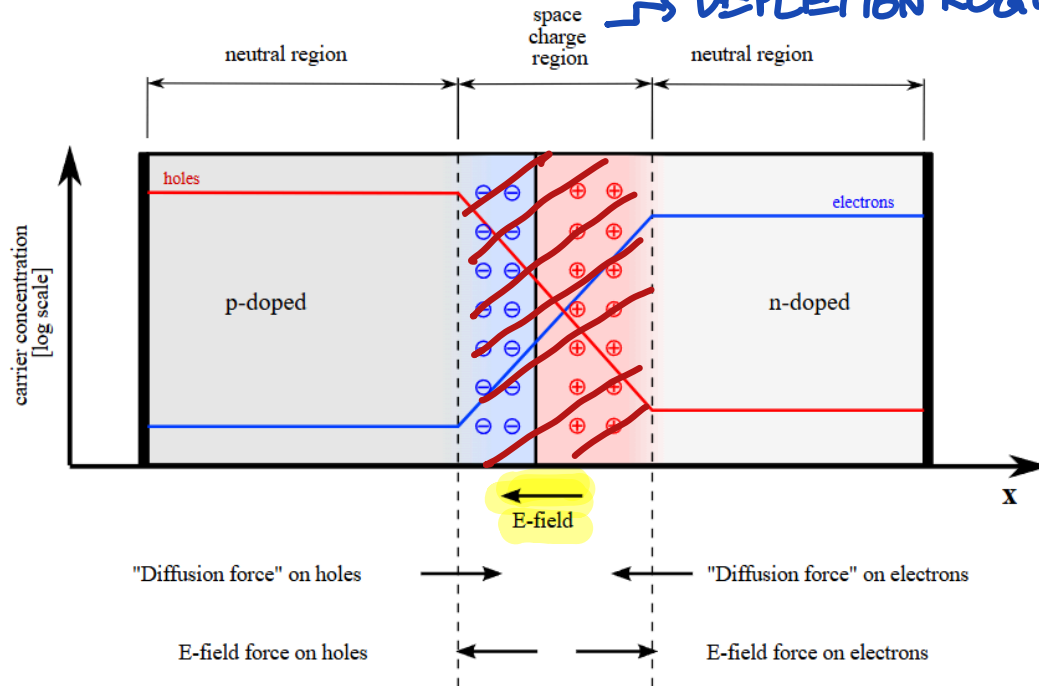


Basic Structure of electronic device

PN Junction - at room temperature

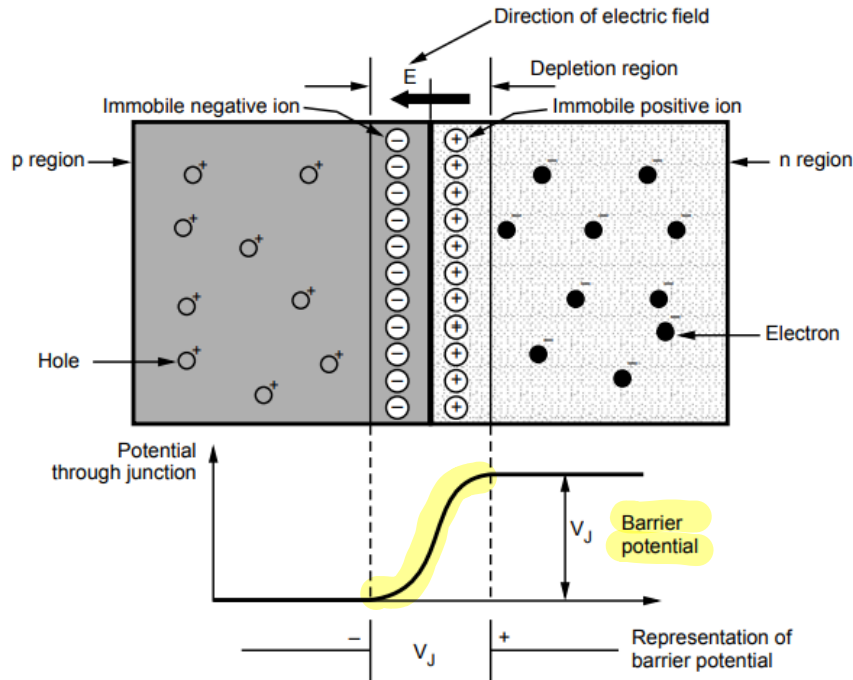
$\approx 25^{\circ}\text{C}$

→ DEPLETION REGION



Basic Structure of electronic device

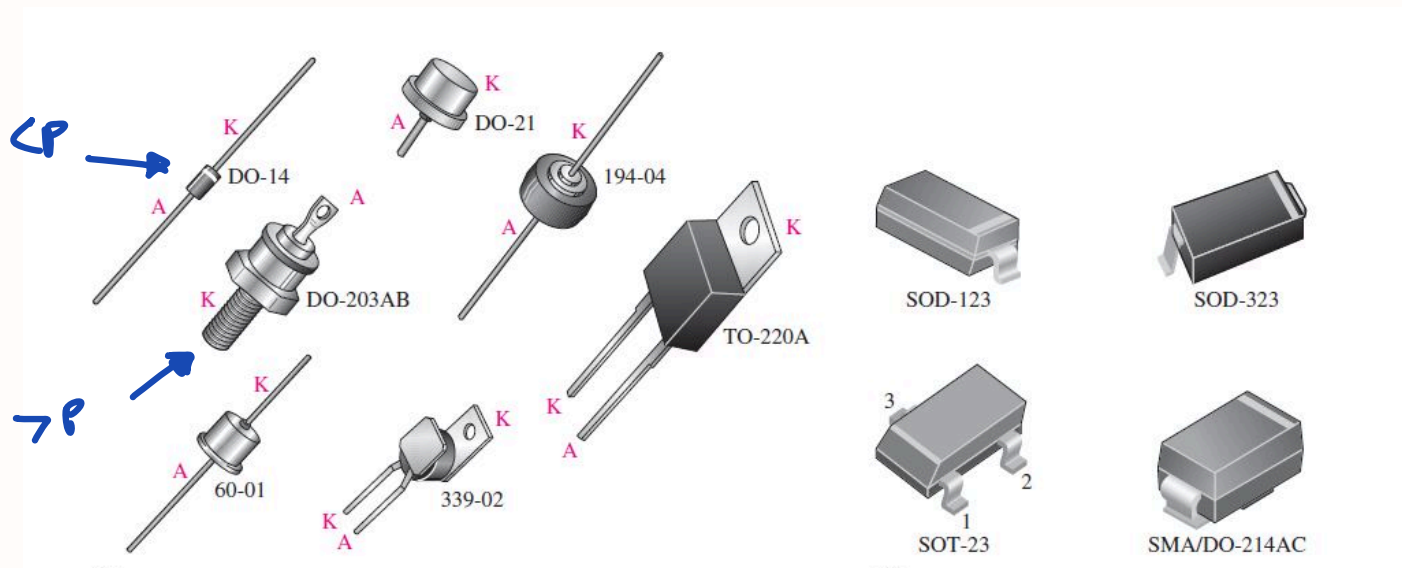
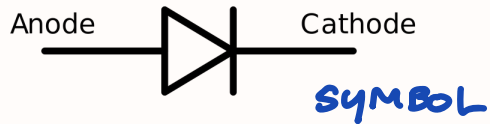
PN Junction - unbiased



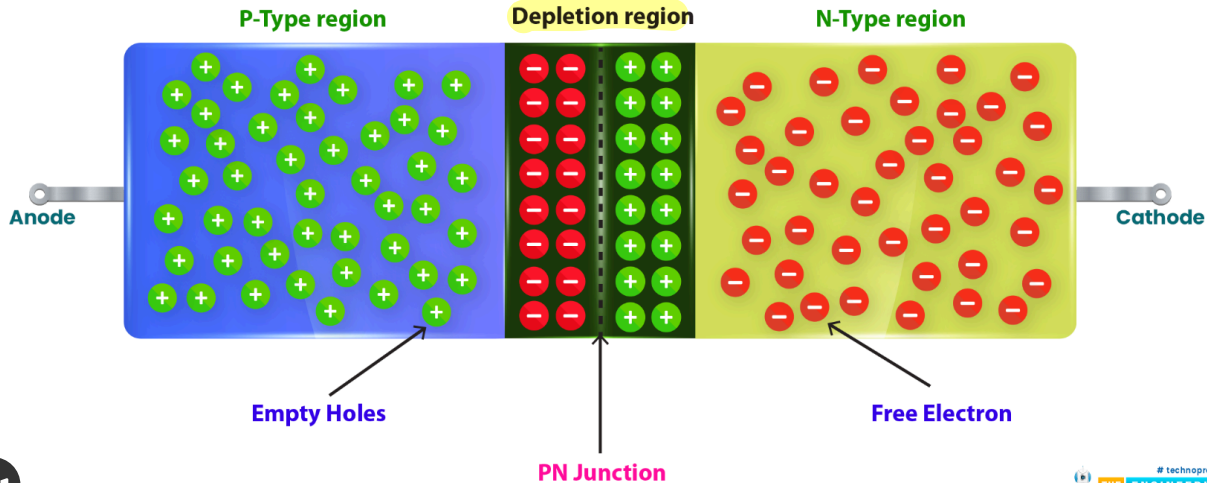
PN junction

- Electric field is generated from the diffusion of carriers
- Electric field creates potential across the barrier
- Depletion layer (or space-charge region) is the area without carriers due to the recombination of free electrons and hole
- Electric field causes barrier potential, which prohibits major carriers to flow to the other side
- Approx. barrier potential of Germanium (Ge) is 0.3 V and 0.7 V for Silicon (Si)

Diode - the simplest electronic device

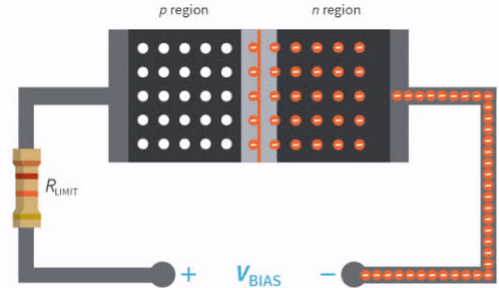
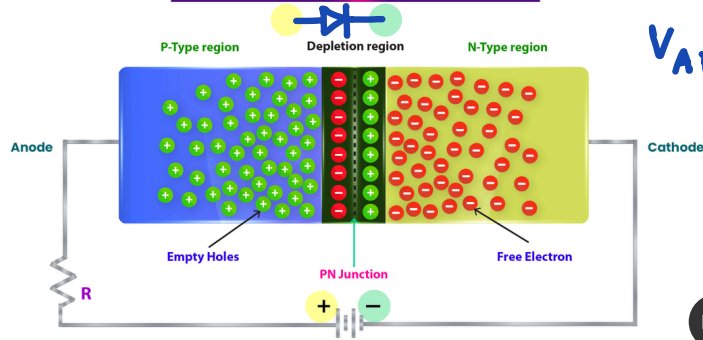


Formation and Properties of Junction Diode

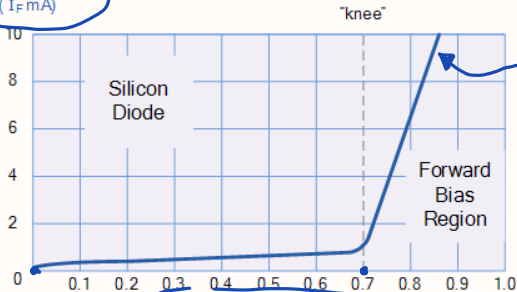


Working principles of diodes

Forward-Biased PN Junction

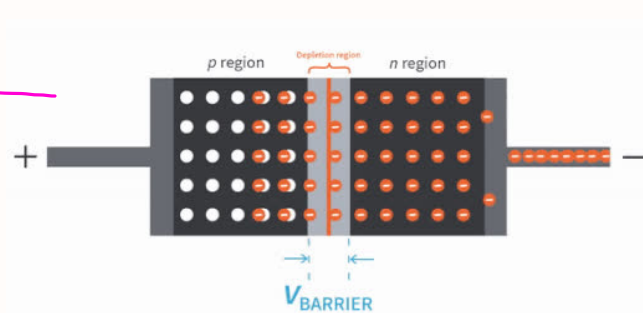


Forward Current (I_F mA)

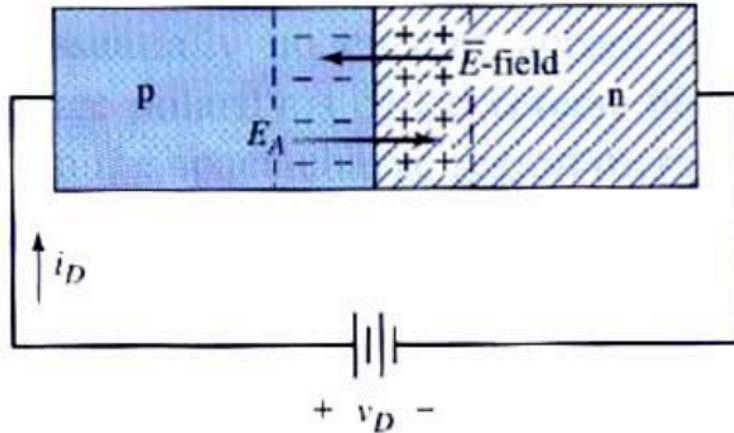


Forward Voltage, (V_F volts)

$$i_D = I_s \left[\exp\left(\frac{v_D}{nV_T}\right) - 1 \right]$$



Working principles of diodes

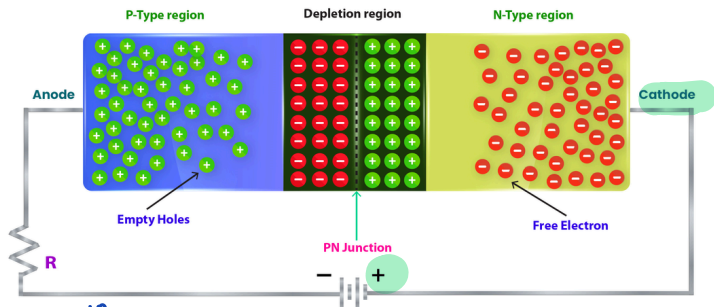


□ Forward-biased PN junction

- Net E-field is less than at the equilibrium, which results in narrower space-charge region and stirs the balance between diffusion and E-field force
- Majority carriers diffuse into the opposite side as long as V_D is applied and create current through the junction just like lowering a dam wall to let lots of water flows

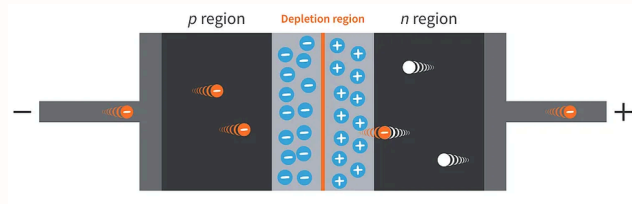
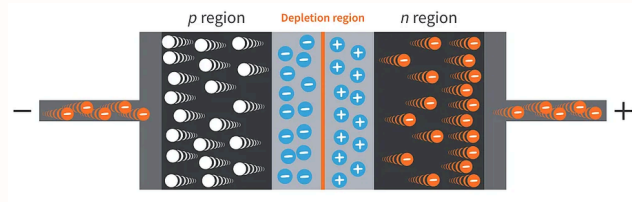
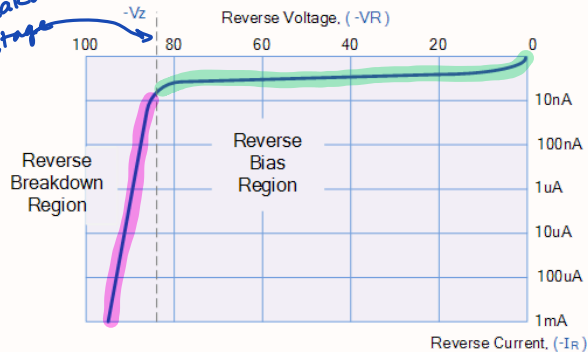
Working principles of diodes

Reverse-Biased PN Junction

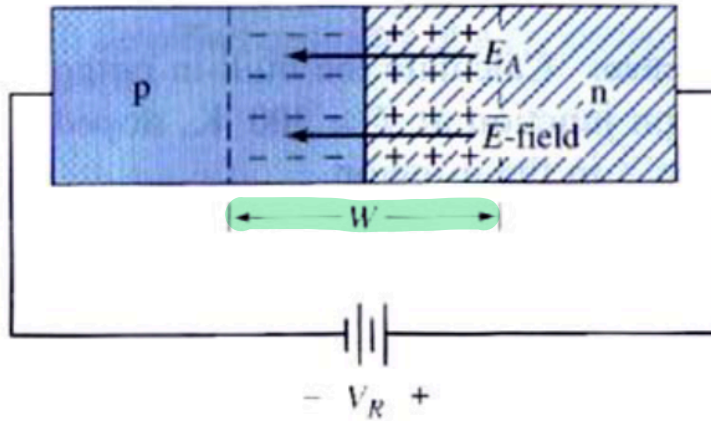


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Breakdown
voltage

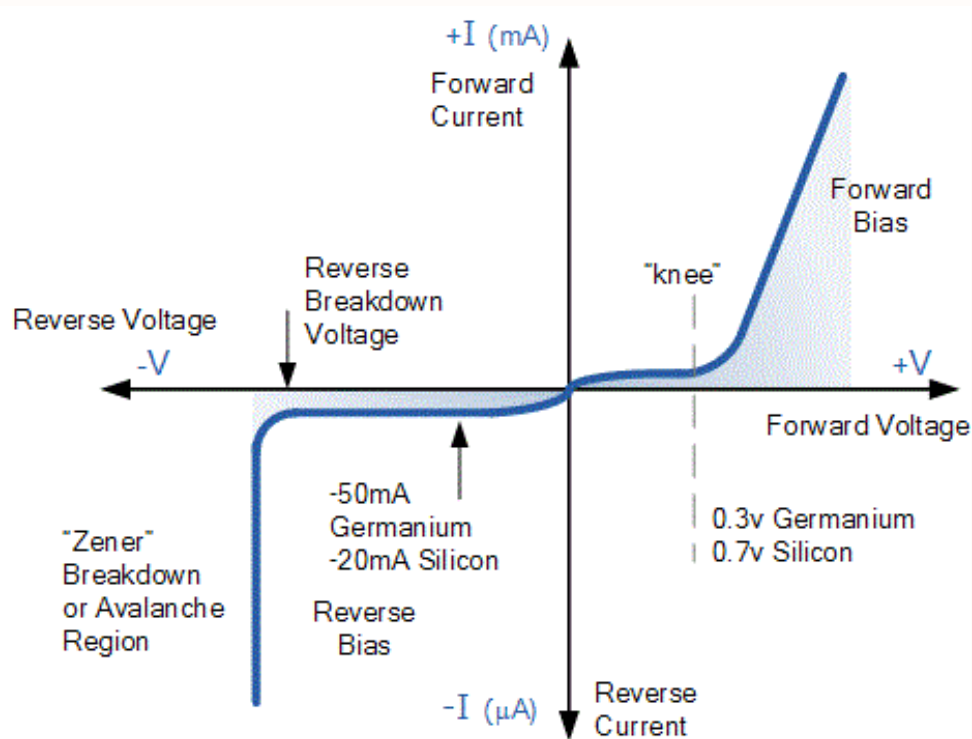


Working principles of diodes

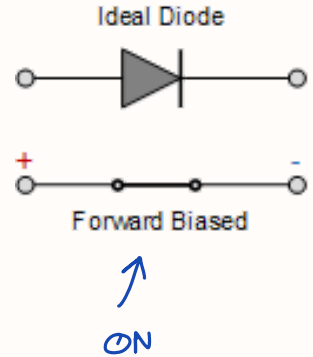
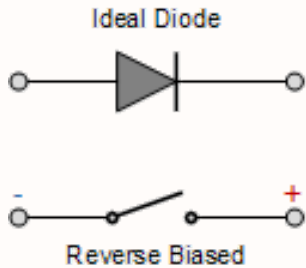
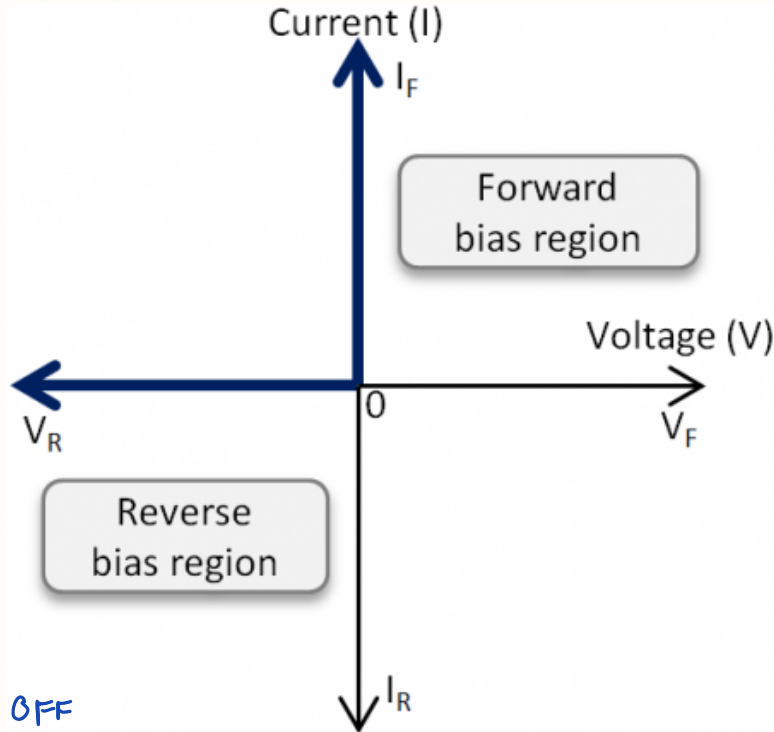


- Reverse-biased PN junction
 - ▣ Electric fields are generated from carrier diffusion and induced from outside supplied voltage V_R (E_A)
 - ▣ Two E-fields have the same direction, which result in larger space-charge region
 - ▣ Higher E-field holds back holes in p-region and electrons in n-region that no current across the junction

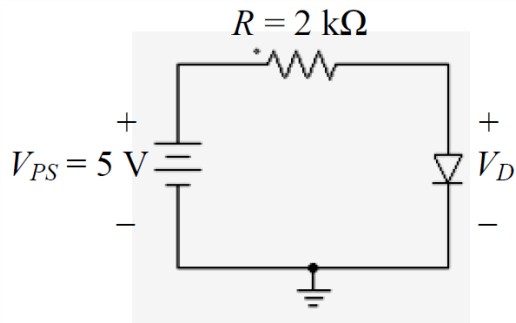
Diode I-V characteristic curve



Ideal diode



Practical diode circuit

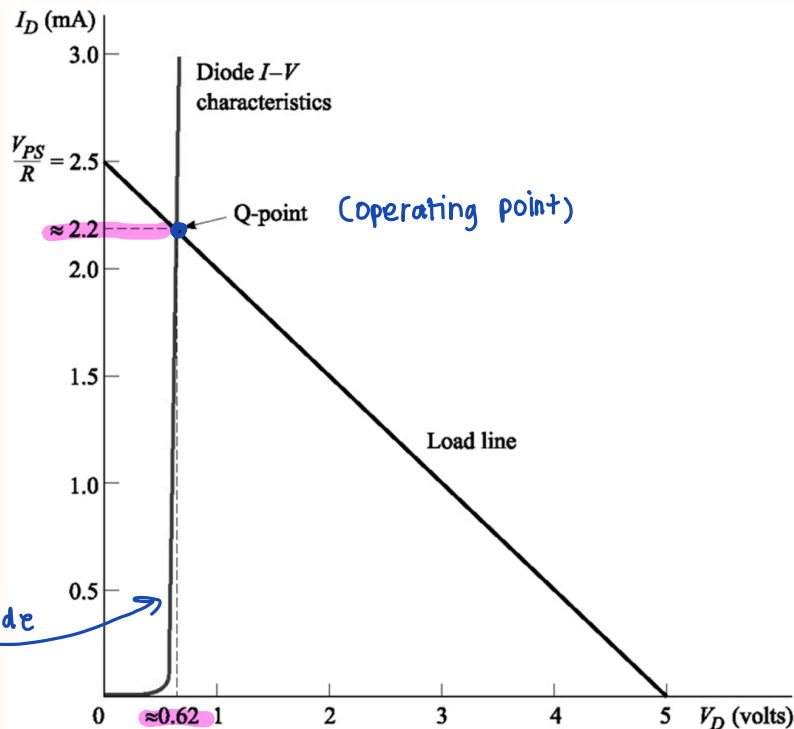


$$V_{PS} = I_D R + V_D$$

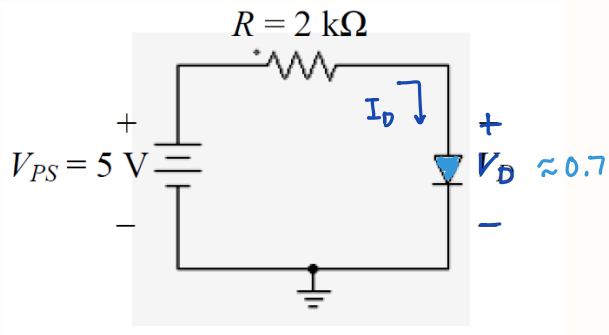
The equation is circled in red, with red asterisks (*) above I_D and V_D . A red arrow points from the text "From the circuit" to the equation.

$$I_D = I_s \left[e^{\left(\frac{V_D}{nV_T} \right)} - 1 \right]$$

The equation is circled in blue. A blue arrow points from the text "From diode" to the equation.



Diode circuit - Non-ideal case



Approximation

$$5 = I_D (2k) + 0.7$$

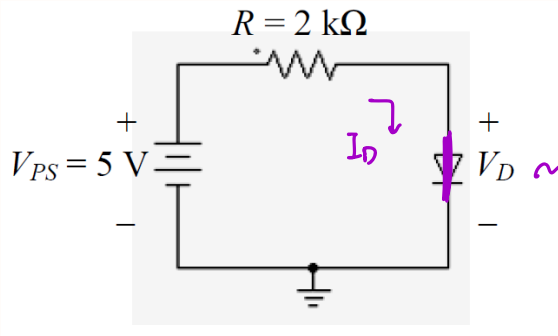
$$I_D = \frac{4.3}{2k} = 2.15\text{ mA}$$

$$I_D = I_s \left[e^{\left(\frac{V_D}{nV_T} \right)} - 1 \right]$$

↙

$$I_D (2k) + V_D = 5$$

Diode circuit - Ideal case

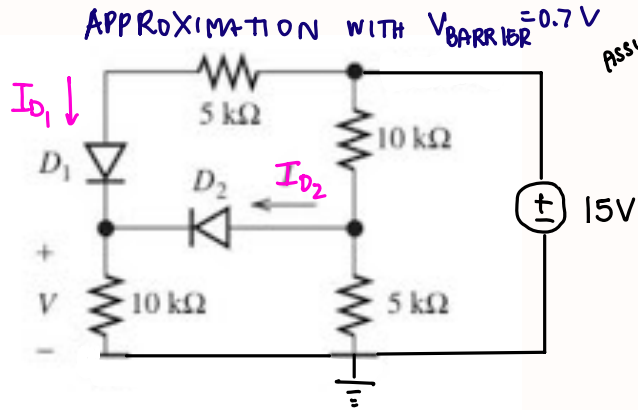


$$V_D = 0$$

$\sim$ ideal

$$I_D = \frac{5}{2k} = 2.5\text{ mA}$$

Non-ideal diode circuit - Example

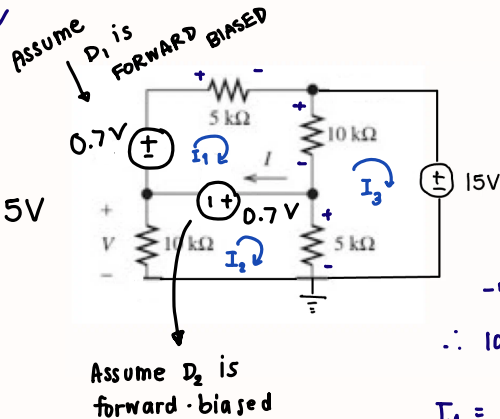


$$I_{D1} = -I_1 = 1.47667 \text{ mA}$$

$\therefore D_1$ is forward-biased

$$I_{D2} = I_1 - I_2 = -0.785 \text{ mA}$$

$\therefore D_2$ is reversed-biased



I_1, I_2, I_3 are mesh currents

$$-0.7 + 5k(I_1) + 10k(I_1 - I_3) + 0.7 = 0$$

$$\therefore 15k(I_1) - 10k(I_3) = 0 \quad \text{--- ①}$$

$$10k(I_2) - 0.7 + 5k(I_2 - I_3) = 0$$

$$\therefore 15k(I_2) - 5k(I_3) = 0.7 \quad \text{--- ②}$$

$$-5k(I_2 - I_3) - 10k(I_1 - I_3) + 15 = 0$$

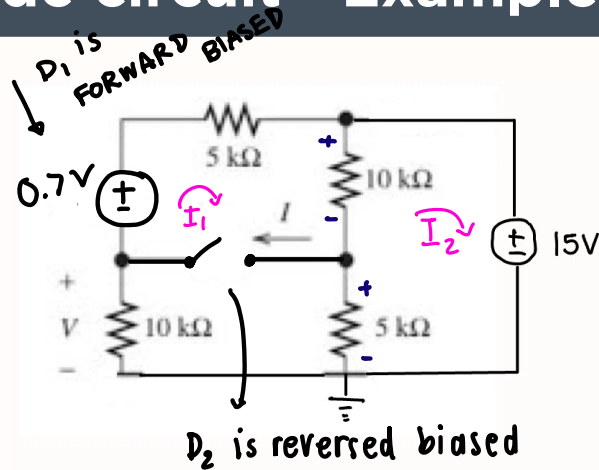
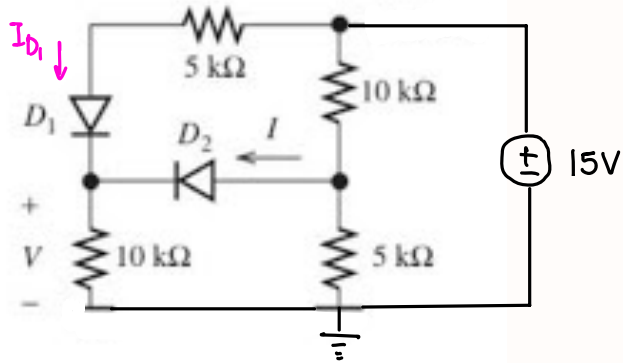
$$\therefore 10k(I_1) + 5k(I_2) - 15k(I_3) = 15 \quad \text{--- ③}$$

$$I_1 = -1.47667 \text{ mA}$$

$$I_2 = -0.691667 \text{ mA}$$

$$I_3 = -2.215 \text{ mA}$$

Non-ideal diode circuit - Example



$$I_1 = -0.953 \text{ mA}$$

$$I_2 = -1.9533 \text{ mA}$$

$$I_{D1} = -I_1 = 0.953 \text{ mA}$$

↳ still forward biased

$$\text{Recheck: } -0.7 + 5 I_1 + 15k(I_1 - I_2) + 10k(I_1) = 0$$

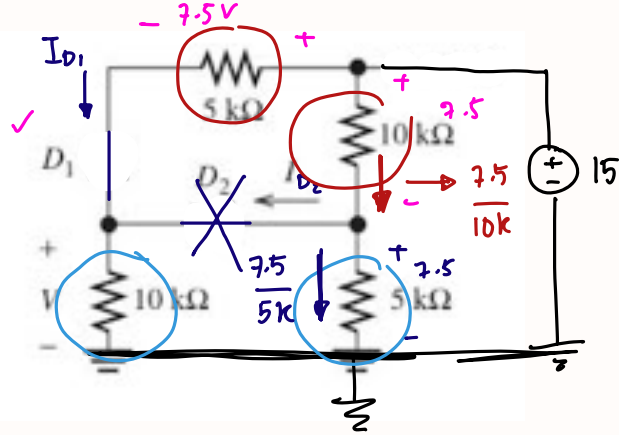
$$\therefore 30k(I_1) - 15k(I_2) = 0.7 \quad \text{--- ①}$$

$$-15k(I_1 - I_2) + 15 = 0$$

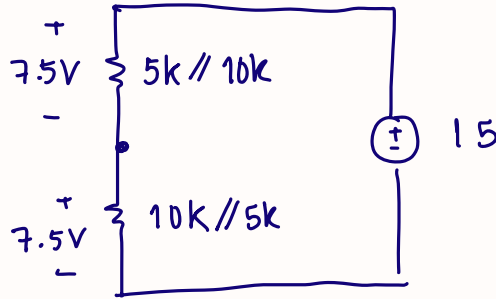
$$15k(I_1) - 15k(I_2) = 15$$

$$\therefore I_1 - I_2 = 1 \text{ mA} \quad \text{--- ②}$$

Ideal diode circuit - Example



$$I_{D1} = \frac{7.5}{5k} = 1.5 \text{ mA}$$



D_1 F.B.

D_2 R.B.

$$I_{D2} + \frac{7.5}{5k} = \frac{7.5}{10k}$$

$$I_{D2} = 0.75 \text{ mA} - 1.5 \text{ mA}$$